Permutation Order

Time limit: 1.00 s
Memory limit: 512 MB

CSES 3397 | Mathematics | https://cses.fi/problemset/task/3397

Let $p(n,k)$ denote the kth permutation (in lexicographical order) of $1 \dots n$. For example, $p(4,1)=[1,2,3,4]$ and $p(4,2)=[1,2,4,3]$.

Your task is to process two types of tests:

1. Given n and k, find $p(n,k)$
2. Given n and $p(n,k)$, find k

Input

The first line has an integer t: the number of tests.

Each test is either "1 n k" or "2 n $p(n,k)$".

Output

For each test, print the answer according to the example.

Constraints

- $1 \le t \le 1000$
- $1 \le n \le 20$
- $1 \le k \le n!$

Example

Input

```
6
1 4 1
1 4 2
2 4 1 2 3 4
2 4 1 2 4 3
1 5 42
2 5 2 4 5 3 1
```

Output

```
1 2 3 4
1 2 4 3
1
2
2 4 5 3 1
42
```